

# MANOJ KHATRI

Backend Software Engineer

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## PROFESSIONAL SUMMARY

Performance-driven Backend Software Engineer with 3+ years of expertise designing and deploying resilient RESTful APIs, distributed microservices, and asynchronous event architectures. Specialized in Python, Django REST Framework, PostgreSQL query optimization, Celery, and Redis caching. Proven record of reducing API latency by 42% and architecting systems serving 100K+ monthly active users with 99.9% uptime.

## EXPERIENCE

Backend Software Engineer | 2023-08 - Present

LightCode Technologies - Kathmandu, Nepal

- Architected high-throughput Django REST API processing 2.4M requests monthly with 99.95% service uptime and sub-120ms p95 latency.
- Optimized complex PostgreSQL database queries and indexes, reducing p99 database query execution times by 44% across 18 core endpoints.
- Engineered asynchronous background job pipeline utilizing Celery, RabbitMQ, and Redis to process 50,000+ daily notifications and transactional invoices.
- Implemented automated CI/CD deployment pipelines with Docker and GitHub Actions, slashing release deployment cycle time from 45 minutes to 7 minutes.

Junior Python Developer | 2022-01 - 2023-07

FinTech Solutions Nepal - Lalitpur, Nepal

- Developed secure payment gateway integrations handling over \$450,000 in monthly digital wallet transactions across eSewa and Khalti APIs.
- Refactored legacy REST endpoints into modular service layer architecture, improving test coverage from 41% to 88% using PyTest.
- Collaborated with frontend engineering teams to design OpenAPI/Swagger contracts and unified error payload standards.

## TECHNICAL SKILLS

Backend & Frameworks: Python, Django, Django REST Framework, FastAPI, Flask, Celery, Asynchronous Programming

Databases & Caching: PostgreSQL, Redis, MySQL, SQLite, Database Indexing, Query Optimization, Transaction Isolation

DevOps & Cloud Tools: Docker, Docker Compose, Git, GitHub Actions, Linux Server

Administration, Nginx, Gunicorn, RESTful APIs, JWT Auth

## KEY PROJECTS

Enterprise Hospital Management System (HMS) | 2024-01 - 2024-09

Production Open Source - Kathmandu, Nepal

- Engineered comprehensive clinical records API managing 12,000+ patient records with role-based access control (RBAC) and audit logging.
- Reduced appointment double-booking error rate to zero by enforcing distributed Redis locks during concurrent checkout workflows.

High-Concurrency Real-Time Analytics Pipeline | 2023-10 - 2024-03

#### Independent Project

- Engineered event ingestion microservice handling 4,500 events/second using FastAPI, Redis Streams, and PostgreSQL partitioned tables.

### **EDUCATION**

Bachelor of Science in Computer Science & Information Technology (B.Sc. CSIT) | 2019-09 - 2023-09

Tribhuvan University - Kathmandu, Nepal

- Core Coursework: Data Structures & Algorithms, Database Management Systems, Distributed Systems, Software Engineering, Operating Systems.

### **CERTIFICATIONS**

Python Certified Associate Programmer (PCAP) | 2023-05 - Present

Python Institute

- Verified advanced proficiency in object-oriented programming, data structures, and standard library internals.